

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2(a)    | <p><i>too small to see</i></p> <p>35 <math>\mu\text{m}</math> is 0.035 mm (smaller than 0.05 mm)</p> <p><b>or</b></p> <p>0.05 mm is 50 <math>\mu\text{m}</math> (larger than 35 <math>\mu\text{m}</math>) ;</p>                                                                                                                                                                                                                                                                                                                                                                                                    | <b>1</b> |
| 2(b)    | <p><i>in context of light microscope</i></p> <p><i>two from:</i></p> <ol style="list-style-type: none"> <li>1 resolution / resolving power, poor / not high enough / AW ;</li> <li>2 ability to distinguish between two points not high enough / AW ;</li> <li>3 ref. to limit of resolution ;<br/>e.g. organelles smaller than, 0.2 <math>\mu\text{m}</math> / 200 nm, not visible<br/><b>A</b> organelles smaller than 200 nm (accept range 100–300 nm)</li> <li>4 organelle too small to interfere with light waves ;<br/><b>A</b> wavelength of light too long</li> </ol> <p><b>I</b> organelles are small</p> | <b>2</b> |
| 2(c)(i) | ester ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>1</b> |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(c)(ii)  | <p><u>facilitated diffusion</u> ;</p> <p><i>plus two from:</i></p> <p>(diffusion) through (membrane), transport / channel / carrier, protein ;      <b>A</b> pore<br/> I transmembrane / integral, protein</p> <p>high to low concentration <b>or</b> down, concentration / diffusion, gradient ;</p> <p>passive ;      <b>A</b> no, ATP / (metabolic) energy, required</p> <p><i>if active transport incorrectly stated allow ecf</i></p> <p>through (membrane), transport / carrier, protein ;      <b>R</b> channel protein</p> <p>low to high concentration / against a concentration gradient / requires ATP / requires (metabolic) energy ;</p>                                                                                                                                                                                                                                                                                                                                       | 3     |
| 2(c)(iii) | <p><i>four from:</i></p> <ol style="list-style-type: none"> <li>1 ref. to hormones, released into / arrive at adipocyte in, blood stream ;<br/> <b>A</b> released by another, cell / tissue (elsewhere in the body)</li> <li>2 hormone is, (chemical) signal / signalling molecule / messenger molecule ;</li> <li>3 (adipocyte is) target cell ;      <b>A</b> (adipose is) target tissue</li> <li>4 (hormone) binds to / AW, receptor ;<br/> <i>in context of</i> on cell surface membrane of adipocyte</li> <li>5 ref.to receptor (shape) complementary to hormone ;<br/> <b>A</b> specific receptor for hormone</li> <li>6 (binding) triggers / stimulates, reactions within the cell ;<br/> <b>A</b> cascade of reactions<br/> I binding causes a reaction to occur</li> <li>7 cell signalling results in a <u>response</u>, qualified ;<br/> e.g. triglyceride breakdown<br/> fatty acids (and glycerol) produced<br/> fatty acids leave, the cell / via transport protein</li> </ol> | 4     |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                        | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2(d)     | <p><i>any one valid suggestion:</i></p> <p><i>e.g.</i><br/> do not have mitochondria / mitochondria required (to produce ATP) ;<br/> do not have nucleus (so no mRNA produced) ;<br/> no mRNA for enzyme synthesis ;<br/> (required) enzymes not present ;<br/> impermeable to fatty acids / AW ;<br/> have lost / do not have, the required organelles ;</p> | 1     |

| Question | Answer | Marks |
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|----------|--------|-------|